

Pointing the need for broader view on AI

Human-Level Intelligence or Animal-Like Abilities?

Communications of the ACM, Oct 2018

Adnan Darwiche

<https://cacm.acm.org/magazines/2018/10/231373-human-level-intelligence-or-animal-like-abilities/fulltext>

“...We need a new generation of AI researchers who are well versed in and appreciate **classical AI, machine learning**, and computer science more broadly while also being informed about AI history...”

Today: Adversarial Examples

What are these? More examples in various domains

Why do they exist?

How to generate adversarial examples?

(techniques also used for adversarial training, logic, etc...discussed in later lectures)

“Adversarial examples are inputs to machine learning models that an attacker has intentionally designed to cause the model to make a mistake”

(Goodfellow et al 2017)

Adversarial Examples [from lecture 1]

Noisy attack: vision system thinks we now have a gibbon...



x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$
“gibbon”
99.3 % confidence

Explaining and Harnessing Adversarial Examples, ICLR '15

Tape pieces make network predict a 45mph sign



Robust Physical-World Attacks on Deep Learning Visual Classification, CVPR'18

Self-driving car: in each picture one of the 3 networks makes a mistake...



DRV_C1: right

DRV_C2: right

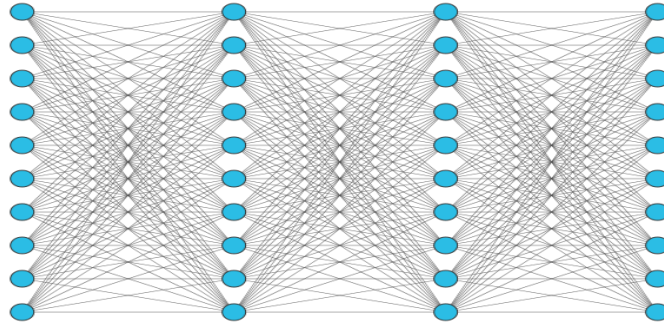
DRV_C3: right

DeepXplore: Automated Whitebox Testing of Deep Learning Systems, SOSP'17

Adversarial Geometric Perturbations



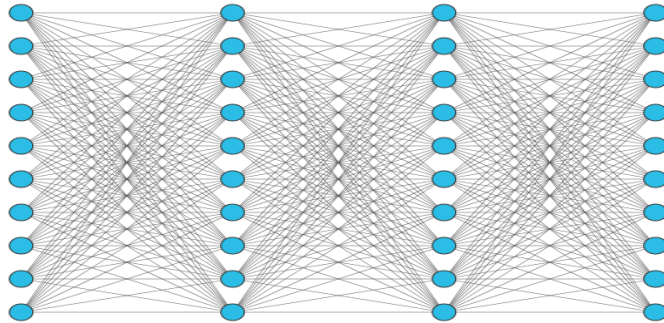
I_o



7

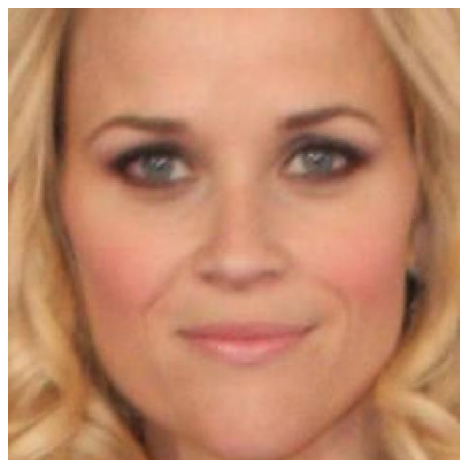


$I = \text{rotate}(I_o, -35)$



3

Adversarial Examples (more)

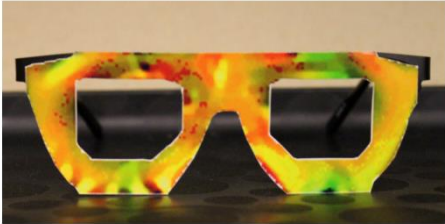


Reese Witherspoon

Russel Crowe

Real World Impersonation/Dodging Attacks

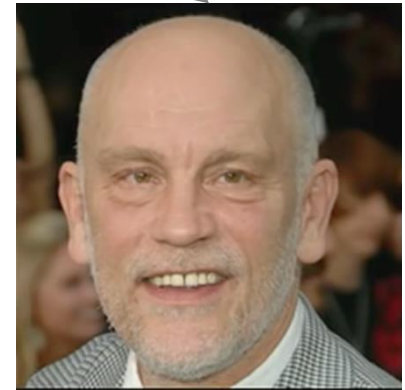
Real glasses



Lujo Bauer



=

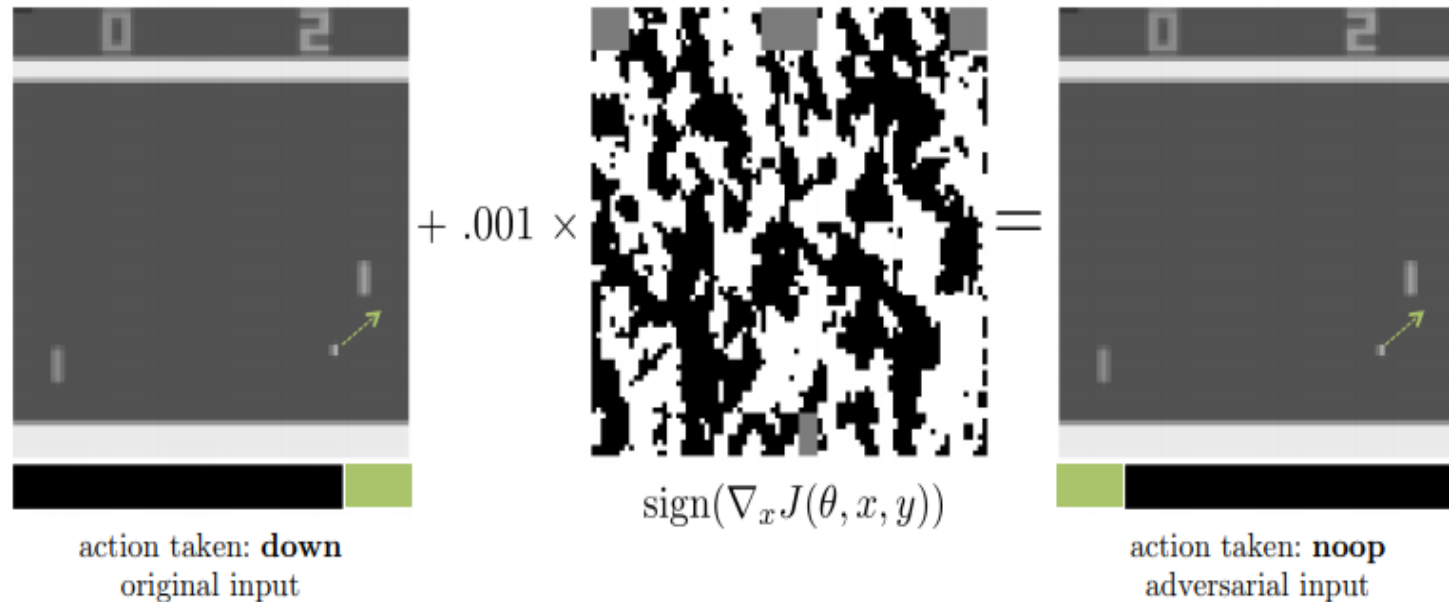


John Malkovich

100% success



Adversarial Examples in Reinforcement Learning



An agent (Deep Q Network) plays the game by selecting actions from a given state (image) that the game produces.

An attacker can perturb the image slightly so that the DQN agent chooses the wrong action: here, it wrongly picks noop (do nothing) in the right image, instead of moving the paddle down (left image).

Adversarial Examples in NLP

Article: Super Bowl 50

Paragraph: *“Peyton Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver’s Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had jersey number 37 in Champ Bowl XXXIV.”*

Question: *“What is the name of the quarterback who was 38 in Super Bowl XXXIII?”*

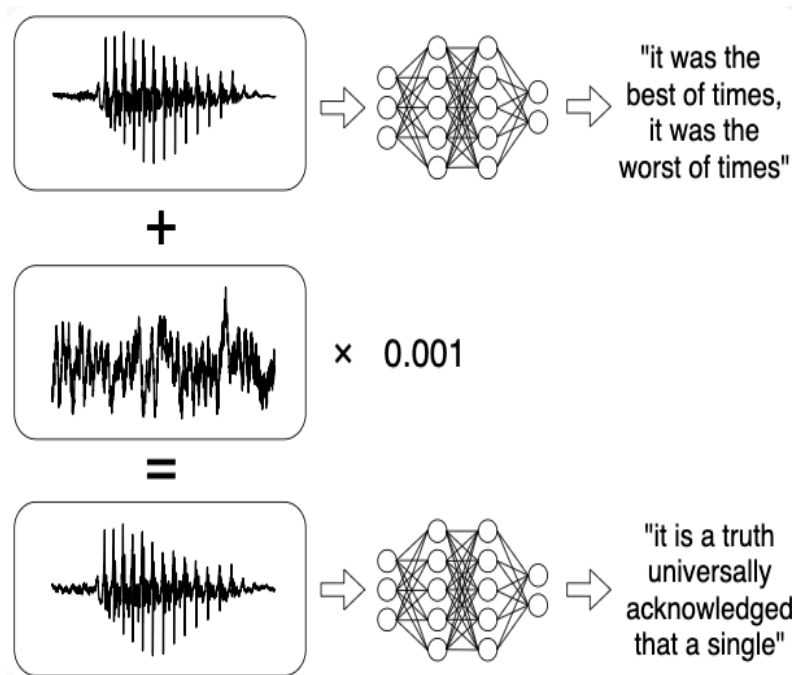
Original Prediction: John Elway

Prediction under adversary: Jeff Dean

The Ensemble model is fooled by the addition of an adversarial distracting sentence in blue.

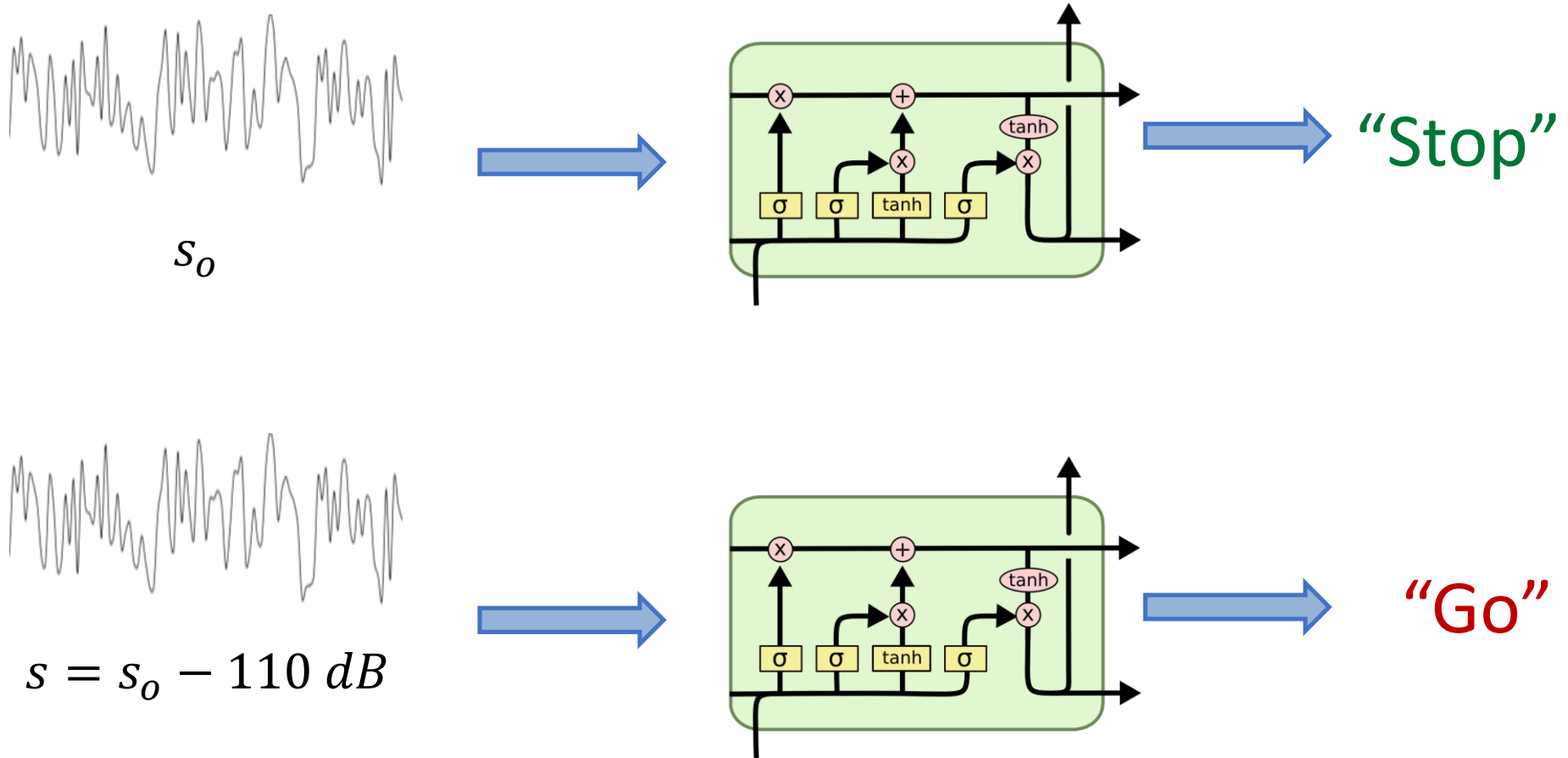
Adversarial Examples in Audio Processing: Speech to Text

An attack on DeepSpeech:



Adding small noise to the input audio makes the network transcribe any arbitrary phrase

Adversarial Examples in Audio Processing: Text classification



Adding small noise to the input audio changes the classification of the output

Adversarial Examples: Some History

2006: Deep learning models gain renewed interest

2012: Multiple works showed that deep networks can achieve near-human performance (sometimes even better)

2013: Research in understanding neural networks behavior becomes critical with society implications beyond computer science

2014: While trying to understand decision making in neural networks, Szegedy et al. discovered adversarial examples

2015-on: Finding adversarial examples and proving their absence becomes an active research area...

Robustness

Robustness: A network is **robust** if it returns correct output on all inputs

Impractical: the input space is too large to be covered

Local Robustness (informal): A learning model is **locally-robust** if it returns the correct output on inputs *similar* to inputs in the training set

This was believed to be evident by having high accuracy on the test set

Why is High Accuracy Not Enough?

Inputs in the training and test set are taken from a given distribution

Neural networks aim to achieve **high accuracy** on test sets drawn **from the given distribution**

There are still many similar inputs that are never tested (and have low-probability for the given distribution)